

The Geometric Mean

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Introduction

The geometric mean is the arithmetic mean on the log scale, back-transformed to the original scale. It is the appropriate summary for variables that combine multiplicatively rather than additively: antibody titres, concentrations, growth rates, investment returns. For log-normal data, the geometric mean is the median and is thus a natural measure of typicality.

Prerequisites

The reader should know the arithmetic mean and be comfortable with natural logarithms.

Theory

For positive values $x_1, \dots, x_n$:

$$\text{GM}(x) = \left(\prod x_i \right)^{1/n} = \exp\left(\frac{1}{n} \sum \log x_i\right).$$

The second form is numerically stable and matches how the geometric mean is computed in practice. Key properties:

- $\text{GM}(x) \leq \bar{x}$, with equality only when all x_i are equal (AM-GM inequality, a consequence of Jensen).
- For log-normal data, the geometric mean equals the population median.
- The ratio of the geometric mean to the arithmetic mean is a simple measure of skewness.

Assumptions

- All values must be strictly positive; a single zero or negative value renders the geometric mean undefined.
- For inference (confidence intervals), log-normality of the data is typically assumed.

R Implementation

```
library(psych)

set.seed(2026)
titres <- 2^sample(4:12, 50, replace = TRUE) # antibody titres in 2^n form

mean(titres)
median(titres)
psych::geometric.mean(titres)
exp(mean(log(titres)))

# 95% CI for the geometric mean
n <- length(titres)
log_x <- log(titres)
```

```
log_mean <- mean(log_x)
log_se   <- sd(log_x) / sqrt(n)
exp(log_mean + c(-1, 1) * qt(0.975, n - 1) * log_se)
```

Output & Results

```
[1] 1157      # arithmetic mean
[1] 512       # median
[1] 588       # geometric mean
[1] 588       # (equivalent computation)

[1] 441 782   # 95% CI for GM
```

The geometric mean (588) sits close to the median (512) and well below the arithmetic mean (1157). This is typical of log-normal or multiplicative data: the arithmetic mean is pulled up by the upper tail.

Interpretation

Report the geometric mean for titres, viral loads, drug concentrations, and any other multiplicatively combining variable. In a paper: “the geometric mean antibody titre was 588 (95% CI 441-782)”.

Practical Tips

- Non-positive values require imputation or exclusion; adding a small constant (e.g., 0.5) before log is common but distorts small values.
- The geometric mean of ratios is the exponent of the mean of log-ratios; reporting log-scale effects is cleaner.
- For growth rates, the geometric mean equals the arithmetic mean of $(1 + r_i)$ values minus 1.
- In meta-analysis of ratios, log transformation + normal approximation is the standard pipeline.
- When data are centred around 1 (e.g., hazard ratios), the difference between AM and GM is small.

For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren’t.
- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

See also — labs in this chapter

- Descriptive statistics and Table 1